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option (b)

4. $P_{N_2} = 0.76$ atm

$K_H = 7.6 \times 10^4$

x = ?

$P_{N_2} = K_H \cdot x$

$0.76 = 7.6 \times 10^4 \times x$

$\therefore x = \frac{0.76}{7.6 \times 10^4} = 1 \times 10^{-5}$ option (d)

5. $K_H = 8 \times 10_4$

$\left(x_{N_2}\right)_{\text{in air}} = 0.5$

Total pressure = 4 atm

pressure Partial pressure of nitrogen
= mole fraction × total pressure
$0.5 \times 4 = 2$

$\left(P_{N_2}\right) = K_H \times$ mole fraction of N_2 in solution

$$2 = 8 \times 10^4 \times \frac{\text{number of moles of nitrogen}}{\text{Total number of moles}}$$

$$\frac{10 + \text{no.of moles of } N_2}{\text{no. of moles of } N_2} = \frac{8 \times 10^4}{2}$$

$$\frac{10}{\text{no. of moles of } N_2} + 1 = 4 \times 10^4$$

$$\frac{10}{\text{no. of moles of } N_2} = 40000 - 1$$

$$\therefore \text{no. of moles of } N_2 = \frac{10}{39999}$$

$= 2.5 \times 10^{-4}$ option (d)

6 for an ideal solution,

$\Delta S_{mix} \neq 0$; Hence $\Delta G_{mix} \neq 0$

$\therefore$ incorrect is $\Delta G_{mix} = 0$ (option d)

7. Carbon dioxide ; most stable gas and has lowest value of Henrys Law constant.

8. $P_{total} = P_1 + P_2$

$= P_1x_1 + P_2x_2$ $\quad x_1 + x_2 = 1$

$= P_1 (1 - x_2) + P_2x_2$ $\quad x_1 = 1 - x_2$

$= P_1 - P_1x_2 + P_2x_2$

$= P_1 - x_2 (P_1 - P_2)$

option (c)

9. $\pi = CRT$

$$\pi = \frac{n}{V} RT$$

$\pi V = nRT$ option (b)

10. Ethanol and water

option (d)

11. Given, $(K_H)_A = x$

$(K_H)_B = y$

$\frac{x_A}{x_B} = 0.2$

$\left(\frac{x_B}{x_A}\right)_{\text{in solution}} = ?$

$P_A = x\,(x_A)_{\text{in solution}} - (1)$

$P_B = y\,(x_B)_{\text{in solution}} - (2)$

$$\left(\frac{x_B}{x_A}\right)_{\text{in solution}} = \frac{P_B}{P_A} \times \frac{x}{y}$$

$$= \frac{x_B}{x_A} \times \frac{x}{y}$$

$$= \frac{1}{0.2} \times \frac{x}{y} = \frac{5x}{y} \text{ (option d)}$$

12. $\frac{\Delta P}{P^o} = \frac{n_2}{n_1}$

$W_2 = 6.5$ g

$W_1 = 100$ g

$K_b = 0.52$

$\frac{\Delta P}{P^o} = \frac{W_2M_1}{M_2W_1}$

$\frac{760 - 732}{760} = \frac{6.5 \times 18}{M_2 \times 100}$

$\therefore M_2 = 31.75$

$\Delta T_b = K_b \cdot m$